Drashti Ankurkumar Bhavsar

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Available full-time for Spring 2027 & summer-27 Co-op — January–August 2027

EDUCATION

Northeastern University, Boston, MA **2025 - Expected December 2027**

MS in Computer Science

Pandit Deendayal Energy University, India **May 2024**

Bachelor of Technology in Computer Engineering ***3.77/4.0***

Coursework: Machine Learning, Deep Learning, Data Mining, Statistical Analysis, NLP, Big Data Analytics

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, JavaScript, C++
ML/DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LightGBM, Transformers, Fine-tuning
Core ML & CV: Feature Engineering, Object Detection, Image Processing
NLP/GenAI: LangChain, LangGraph, RAG, NLTK, spaCy, Prompt Engineering, Vector Databases
Data Engineering & BI: PySpark, Apache Kafka, Hadoop, ETL Pipelines, Data Warehousing, Tableau, Power BI
MLOps: Docker, Kubernetes, MLflow, FastAPI, Model Deployment, CI/CD, A/B Testing, Evidently AI, Streamlit
Cloud & Tools: AWS (EC2, S3, SageMaker), GCP, Azure, Git, Airflow

EXPERIENCE

LOTI AI Inc. **August 2024 – July 2025**

Machine Learning Engineer Intern | OpenCV, BGE, Qwen, Scrum, Data Science, Jira *Seattle, WA*

- Built and deployed a multi-scale ML system in PyTorch for real-time content classification, improving decision accuracy to 87.3% and outperforming production baselines by 12%.
- Designed end-to-end ML pipelines (feature engineering → training → inference → monitoring) supporting real-time decision-making for 10K+ daily requests with 99% uptime.
- Improved 75% F1-score and 97% recall for AI prompt-routing pipeline using hyperparameter tuning and cross-validation.

IIT Roorkee - Google ExploreCSR **January 2024 – June 2024**

Junior Research Fellow | PyTorch, TensorFlow, OpenCV, NumPy, Pandas, SciPy *Uttarakhand, India*

- Designed hierarchical Deep Reinforcement Learning models in CARLA simulator for autonomous vehicle navigation, training agent with depth cameras and obstacle detection modules.
- Improved decision-making accuracy by 25% in dynamic environments through feature engineering and reward function optimization, solving complex overtaking and lane-changing problems.
- Launched model achieving 94% success rate in simulated urban scenarios with real-time inference under 50ms latency.

Indian Space Research Organisation - ISRO **June 2023 – August 2023**

Data Science Research Intern | Python, Random Forest, LSTM, Forecasting, Classification *Ahmedabad, India*

- Analyzed satellite navigation signal anomalies using time-series modeling to support positioning and fault detection.
- Trained and evaluated Random Forest and LSTM models achieving 98% and 97% accuracy, respectively
- Released real-time L5-signal classification model using FastAPI with 23% improved positioning accuracy; implemented internal network accessibility through ngrok tunneling for production monitoring.

PROJECTS **GitHub**

Intelligent Traffic-Aware Delivery Optimization | *Python, A-star, Dijkstra, Q-Learning, DQN*

- Built a traffic-aware routing system comparing A-star, Dijkstra, Greedy Best-First Search and TSP heuristics across dynamic delivery scenarios.
- Developed Q-learning and DQN agents to adapt routing decisions based on traffic conditions and evaluated total travel time, customer wait time, completed deliveries and runtime.

FIFA 2026 World Cup Predictor | *XGBoost, Random Forest, FastAPI, Next.js*

- Trained an ensemble of three XGBoost models and a Random Forest on 35,304 matches using 97 features; achieved 0.826 log loss and 0.018 expected calibration error.
- Identified a 7% bias in Elo-only predictions, combined multiple rating sources, and ran 10,000 Monte Carlo simulations with Poisson scorelines; built a FastAPI and Next.js application.

Patent Analytics research | *SQL, LLM, R, Statistical Analysis, Hypothesis Testing, A/B Testing, Forecasting*

- Analyzed more than 10,000 patent records to identify trends distinguishing AI and non-AI innovations, developing a classification model with 92% accuracy.
- Automated data collection and transformation workflows, reducing manual ETL effort by 50%, and developed dashboards for communicating five-year technology trends.

PUBLICATIONS **Google Scholar**

- Health Misinformation Detection in Hidradenitis Suppurativa Communities Using Multi-Model LLM Ensembles
- Milk Quality Prediction Using Machine Learning. ***(EBSCO)***